

Connect Four Plus Variant Design

A phone-friendly pass-and-play rule design for powerups, fog of war, gravity, tokens, and other strategic Connect Four variants.

Core thesis

Once a Connect Four-style game includes hidden powerups, private hands, fog of war, or secret objectives, normal same-screen pass-and-play needs a privacy-preserving turn flow. Otherwise one player can see the other player's private information when the phone is passed.

The goal of these rules is to keep the game playable, fun, fair for both players, and strategically meaningful instead of chaotic.

Core pass-and-play privacy system

For any mode with hidden information, each turn should use three phases.

Phase	What the app shows	Why it matters
1. Handoff screen	A neutral screen: "Pass the phone to Player B. Player B, tap to reveal your turn." No board, no powerups, no history that reveals hidden information.	Prevents the previous player from seeing the next player's private state.
2. Private turn reveal	The active player taps Start Turn and sees only what they are allowed to see: their powerups, their fog view, legal moves, and relevant private hints.	Allows hidden hands and asymmetric board views.
3. Commit move	The player chooses and confirms a move. Public effects resolve, then the app returns to the handoff screen.	Makes each move deliberate and preserves secrecy between turns.

Global design principles

- Keep the base goal recognizable: first to connect four horizontally, vertically, or diagonally.
- Limit powerups: start with 1, hold at most 3, use at most 1 per turn.
- Used powerups should become public, but unused powerups stay private.
- Avoid instant unfair wins. Strong effects should have setup, limits, or counterplay.
- Prefer tactical disruption over hard denial. Locking a column for one turn is usually better than skipping a turn.
- Check wins at consistent moments so players can reason about outcomes.

Variant 1: Power Duel

This is the strongest starting mode: normal Connect Four with hidden powerups.

Rules

- Use a standard 7-column by 6-row board.
- Each player starts with 1 hidden random powerup.

- Each player may hold up to 3 powerups.
- On your turn: optionally use 1 powerup, drop 1 piece, resolve effects, check for a win, then hand off.
- After every 4th personal turn, draw 1 hidden powerup unless your hand is full.

Recommended powerups

Powerup	Rule	Balance note
Pop	Remove your own bottom piece from any column. Pieces above fall.	Flexible, but cannot directly remove the opponent's piece.
Lock	Choose a non-full column. Opponent cannot play there next turn.	Cannot lock the only legal column.
Swap Top	Swap the top two pieces in one column.	Limited enough to be tactical without rewriting the board.
Shield	Protect one of your pieces from removal effects.	Creates counterplay against Pop/Bomb-style effects.
Double Drop	Drop two pieces this turn in different columns.	Very strong. Make rare or delayed until later turns.

Variant 2: Draft Duel

A more competitive version with less randomness. Players secretly choose abilities before the game.

- Before the game, each player secretly drafts 3 one-use powerups from the same list.
- Both players may choose the same powers in the simplest version.
- Unused powers remain hidden. Used powers become public.
- On each turn, a player may use at most 1 drafted power before placing a piece.

This mode is fairer than random powerups because players choose a plan. A player can draft an aggressive kit, defensive kit, or control kit.

Kit	Example powers	Play style
Aggressive	Double Drop, Swap Top, Pop	Pushes threats and creates forks.
Defensive	Lock, Shield, Scan	Blocks threats and preserves key pieces.
Control	Lock, Gravity Nudge, Swap Top	Manipulates where the opponent can safely play.

Variant 3: Shadow Connect / Fog of War

Fog of war can be fun, but too much hidden information turns the game into guessing. The player should still be able to reason about the board.

Visibility rules

A piece is visible to you if it meets at least one condition:

- It is your own piece.
- It is directly adjacent to one of your pieces horizontally, vertically, or diagonally.

- It is the top occupied cell of a column.
- It is part of a visible three-in-a-row threat.
- It was placed on the opponent's immediately previous turn.

Hidden occupied spaces should never appear empty. Show them as gray mystery tokens so players know a space is occupied without knowing whose piece it is.

Win checking uses the true board state. When someone wins, reveal the full board and highlight the winning line.

Second-player balance option: give Player 2 one Scan powerup at the start, or give Player 2 slightly wider visibility for their first two turns.

Fog-specific powerups

Powerup	Rule	Caution
Scan Column	Reveal one full column until the end of your turn.	Simple and fair.
Scan Radius	Reveal all hidden pieces within two spaces of a selected cell.	Good for tactical planning.
Decoy	Hide one of your visible pieces again unless it is part of a three-in-a-row threat.	Use sparingly; too much hiding feels unfair.
Signal Jam	Opponent's next Scan reveals only part of the selected area.	Advanced; may frustrate casual players.

Variant 4: Powerups plus Fog

This is the most complex mode. It should be treated as an advanced mode, not the default.

- Use the Shadow Connect visibility rules.
- Players also have hidden powerup hands.
- A player may not target a piece they cannot see unless the powerup is specifically an information power.
- The opponent sees that a powerup was used and sees public effects, but does not see the active player's remaining hand or private board view.

Recommended powerups for this mode: Scan Column, Lock, Shield, and possibly Swap Top. Avoid too many removal or anti-information powers.

Variant 5: Gravity Connect

A visually strong public-information mode. It usually does not need hidden hands.

- Use a 7x7 board or a standard 7x6 board.
- Gravity has a current direction: down, left, right, or up.
- A piece enters from the valid edge and slides as far as possible in the current gravity direction.
- Every 6 total turns, after the active player places a piece and win checks resolve, gravity rotates clockwise.
- Always check for a win before rotating gravity.

The strategic hook is planning for threats that become stronger after the next gravity shift.

Variant 6: Bomb Pieces

Bombs are fun but can become unfair if they erase too much of the board.

- Each player gets 2 bombs for the game.
- Instead of dropping a normal piece, a player may drop a bomb piece.
- When it lands, it destroys orthogonally adjacent pieces: above, below, left, and right.
- It can destroy both players' pieces.
- Pieces above destroyed cells fall down.
- Resolve the explosion and falling pieces before checking for wins.

Do not destroy diagonal pieces. Limit the blast to keep the game readable and strategic.

Variant 7: Wild Pieces

Wild pieces create forks and tactical surprises, but simultaneous wins need a clear rule.

- Each player has 1 neutral wild piece.
- A neutral wild counts as either player's color when checking lines.
- If a wild creates a connect four for both players, the active player wins.
- Wild pieces cannot be removed by Pop, but can be destroyed by Bomb if Bombs are enabled.

Alternative simpler rule: a wild piece counts only for the player who placed it. This is easier for casual play, but less interesting.

Variant 8: Column Locks

One of the easiest and cleanest variants.

- Each player has 2 locks.
- Before placing, choose a non-full column to lock.
- Opponent cannot play in that column on their next turn.
- You must place in a different column after locking.
- You cannot lock the only playable column.
- You cannot lock the same column twice in a row.
- Locks are public and expire automatically.

This creates short-term tactics without making the game feel random.

Variant 9: Shrinking Board

This works best when the shrink schedule is public and predictable.

- Start with a 9-column board.
- After turn 10, remove the leftmost column.
- After turn 18, remove the rightmost column.
- Pieces in removed columns disappear.

- Check for wins before removing a column.
- Warn both players at least 2 turns before a column disappears.

Avoid shrinking too often. Frequent board removal makes the game feel arbitrary.

Variant 10: Connect 5 Mode

A simple, clean strategic variant.

- Use a 9-column by 7-row board.
- First to connect 5 wins.
- Optional: each player receives one Double Drop powerup after turn 8.

Connect 5 reduces early traps and gives players more room to build plans.

Variant 11: Token Hunt

A strong way to let players earn powerups through board play rather than pure randomness.

- At game start, place 3 public power tokens on reachable board cells.
- When a player's piece lands on a token cell, they draw 1 hidden powerup.
- Tokens should appear mostly on lower reachable rows.
- A new token appears 2 turns after one is collected.
- Tokens cannot appear in full columns.
- Powerup locations are public, but drawn powers are private.

This creates a strong tradeoff: block a threat, build your own line, or grab a token.

Variant 12: Hidden Objectives

Hidden objectives can add replayability, but should not replace the main win condition at first.

- Each player secretly receives one objective.
- Completing the objective grants a powerup instead of points.
- Examples: create a diagonal threat, use the center column three times, create two threats at once, or force the opponent into an edge column.
- The objective is revealed only when completed.

This keeps Connect Four as the main game while rewarding alternate strategies.

Variant 13: Puzzle Campaign

A single-player mode that teaches advanced mechanics and gives the app more depth.

- Win in 1 move.
- Win in 2 moves.
- Block all opponent threats.
- Use exactly one powerup.

- Survive 3 turns.
- Create a fork.

Suggested campaign: 10 classic puzzles, 10 powerup puzzles, 10 fog puzzles, 10 gravity puzzles, and 10 expert mixed puzzles.

Variant 14: Simultaneous Planning

More experimental. Both players secretly choose a column, then moves resolve together.

- Player A secretly chooses a column.
- Handoff screen.
- Player B secretly chooses a column.
- Reveal and resolve both moves.
- If both choose the same column, priority alternates by round.

This becomes a prediction game and should be treated as a separate mode.

Privacy requirements by variant

Variant	Needs handoff privacy?	Reason
Classic Connect Four	No	Everything is public.
Power Duel	Yes	Hidden powerup hand.
Draft Duel	Yes	Hidden ability choices.
Shadow Connect / Fog	Yes	Different board views.
Powerups plus Fog	Yes	Hidden hands and hidden board state.
Gravity Connect	No	Fully public state.
Bomb Pieces	Maybe	Only if bombs are hidden before use.
Wild Pieces	Maybe	Only if wild pieces are hidden before use.
Column Locks	Maybe	No if locks are public; yes if lock cards are hidden.
Shrinking Board	No	Public schedule.
Token Hunt	Yes	Tokens are public, drawn powerups are hidden.
Hidden Objectives	Yes	Secret goals.

Recommended polished modes

Mode	Best for	Summary
Power Duel	Default casual mode	Normal Connect Four plus hidden powerups earned every 4th personal turn.
Draft Duel	Competitive mode	Each player secretly drafts 3 one-use powers before the game.
Shadow Connect	Advanced fog mode	Players see partial board information and use Scan powers.
Gravity Connect	Visual/arcade mode	Gravity rotates on a predictable schedule.

Mode	Best for	Summary
Token Hunt	Board-control mode	Players collect public tokens that grant hidden powerups.

Strong default rulebook: Connect Four Plus - Power Duel

Goal

Be the first player to connect four of your pieces horizontally, vertically, or diagonally.

Board

Use a standard 7-column by 6-row board.

Setup

- Each player chooses a color.
- Each player receives 1 hidden random powerup.
- Each player may hold up to 3 powerups.

Turn

- View your private turn screen.
- Optionally use 1 powerup.
- Drop 1 piece into a non-full column.
- Resolve falling, power effects, and win checks.
- If no one wins, end your turn.
- The app hides the board and asks you to pass the phone.

Gaining powerups

After every 4th personal turn, draw 1 hidden powerup unless your hand is full.

Win checking

- Check for a win after all effects and drops resolve.
- If both players connect four on the same turn, the active player wins.
- If the board fills with no winner, the game is a draw.

Privacy

- A player's powerups are visible only on that player's turn.
- After each turn, the app hides the board until the next player taps Start Turn.
- Used powerups are shown publicly in the game log.

Implementation architecture

Build the engine around separate public, true, and private state objects.

State object	Purpose
trueBoardState	The actual board used for win checking and physics.
publicBoardState	What both players are allowed to know. Same as trueBoardState in public modes.
playerOnePrivateState	Player 1 powerups, hidden objectives, and fog visibility.
playerTwoPrivateState	Player 2 powerups, hidden objectives, and fog visibility.
publicGameLog	Used powerups and visible effects.
pendingHandoff	Whether the app should hide the board before the next turn starts.

For normal modes, publicBoardState and trueBoardState are the same. For fog modes, trueBoardState stores the real board and each player gets a filtered board view. For powerup modes, each player has a private hand and the public log records only used powers.

Build order recommendation

- First build Power Duel: it validates whether powerups are fun without complex board visibility.
- Then build Draft Duel: it gives a fair competitive version with less randomness.
- Then build Token Hunt: it adds board-control incentives.
- Then build Shadow Connect: fog of war is more complex and should come after the base engine is solid.
- Finally build Gravity Connect or Bomb/Wild variants as alternate arcade modes.

Direct recommendation: make Power Duel the default, Draft Duel the competitive mode, and Shadow Connect the advanced mode.